PulseIQ HR Analytics

Phase 2 — Data Analytics Project Report

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Track	Data Analytics
Repository	github.com/ShivamSingh1208/Data-Analytics_project
Git branch	main
Git tag	phase-2 — created and pushed to GitHub
Reporting period	April 2025 – September 2026 for budget and KPI source data

Executive summary

Phase 2 extends the PulseIQ HR Analytics warehouse with attendance and leave metrics, attrition metrics and grouped risk summaries, productivity metrics, payroll reconciliation, and department payroll budget analysis. The work uses PostgreSQL tables and SQL views, Python loaders, source-data checks, and manual comparisons. Seven automated ETL tests passed. The work includes KPI target comparisons, budget analysis, and documented payroll reconciliation differences that remain pending data-owner review.

Task status and evidence

Workstream	Status	Evidence / remaining work
Attendance & leave	Implemented; verified	SQL views created. Monthly attendance, department metrics, leave status totals and date checks reviewed.
Attrition	Implemented; verified	891 monthly department records and 430 risk predictions loaded. Group risk reporting suppresses groups under 10.
Productivity	Implemented; verified	16 teams, 38 sprints and 28,868 tasks loaded. Cancelled work excluded from key completion metrics.
Payroll reconciliation	Implemented; caveats documented	18 monthly reference checks completed. Four months differ from the official reference; rejected rows and reasons documented.
Budget & KPI targets	Implemented; validated with caveats	198 budget rows and 180 KPI target rows loaded. Budget source checks passed; KPI comparison view validated. Eighteen task targets are missing.

Verified Results

Attendance, attrition, productivity and payroll

Attendance and leave

Monthly attendance ranged from 92.89% in May 2026 to 95.51% in July 2025. Finance & Accounts recorded 90.82% across 6,076 scheduled days, and HR recorded 92.51%. These are investigation signals, not proof of a cause. The leave warehouse contains 3,439 records; leave-status totals were checked. Seven approved medical-leave records each span 130 days (910 days in total) and should be validated with the source owner before interpretation.

Attrition

The headcount source contains 891 department-month records from January 2020 through September 2026. The risk source contains 430 predictions, including 43 High and 86 Medium predictions in the September 2026 snapshot. Risk predictions are not actual exits and are not presented as certainty. The March 2026 snapshot had only five predictions and is suppressed from group reporting under the minimum-group-size rule.

Productivity

The task warehouse contains 28,868 tasks across 16 teams and 38 sprints. There are 24,139 Done tasks, 898 Cancelled tasks and 999 tasks with Blocked status. Overall on-time completion was 92.86% among completed tasks with a valid on-time result; average valid cycle time was 4.80 days. QA Automation had a 94.01% on-time rate, while Growth Marketing had 91.30%. These differences do not establish why team performance varies.

Payroll reconciliation

The official reference covers 18 months. Four months differ after ETL cleaning: May 2025 (INR 61,813.15), December 2025 (INR 98,327.42), February 2026 (INR 154,811.89), and April 2026 (INR 96,015.00) in employer cost. The May and February records include duplicate employee/run pairs. The December and April records have employee IDs not found in the employee dimension. Rejected rows remain excluded pending data-owner review.

Budget analysis

The 198 department-month budget rows total INR 862,030,000.00. Cleaned payroll employer cost totals INR 861,369,767.38, giving a net variance of –INR 660,232.62 (actual minus budget). Largest net overspends are Data & AI INR 992,768.20 (0.90%), Customer Support INR 482,050.62 (1.01%), and Product & Design INR 453,539.93 (0.65%). Engineering had the largest net underspend at INR 2,054,508.87. Three department-month budget values matched the original CSV.

Quality, Lessons and Next Steps

Testing and quality controls

Automated tests: 7 passed using the existing ETL test suite. Tests cover reversed leave dates, date-key consistency, duplicate identifiers, negative payroll amounts and open-ended validity dates.

Warehouse checks: Source-to-warehouse department/month counts and three manual budget checks were reviewed. The payroll budget view contains 198 rows across 11 departments and 18 months. KPI targets contain 180 rows across 10 departments and 18 months.

KPI comparison results: The view contains 180 department-month rows across 10 departments and 18 months. Attendance met target in 106 rows, overtime in 152, attrition in 121, and task completion in 136 of 162 rows with an available target. Eighteen task-completion targets remain NULL. Attendance target 0.94 was converted to 94%. Overtime share assumes overtime hours divided by scheduled days multiplied by 8 hours; this denominator needs confirmation. Recorded overtime is zero in 126 of 198 attendance department-month rows, so overtime findings need source validation.

Problems and resolutions

- Source budget values included comma separators; the loader removes commas before numeric conversion.
- A pre-existing SQL view had a different column layout; it was safely recreated without deleting payroll or budget data.
- Payroll duplicate pairs caused two monthly reconciliation gaps; the records remain excluded pending data-owner confirmation.
- Missing KPI targets are preserved as NULL instead of being fabricated.

Learning reflection

This phase strengthened my practical understanding of loading source data into PostgreSQL, designing SQL metric views, validating keys and totals, testing ETL rules, and reconciling payroll with an official reference. I also learned to distinguish actual results from model predictions, preserve missing values, and document unresolved issues instead of hiding them.

Next steps

1. Confirm the four monthly payroll reconciliation differences with the data owner and resolve them through an approved process.
2. Confirm the overtime-share denominator and investigate zero-overtime source values.
3. Confirm rejected payroll rows with the data owner and keep missing task targets unavailable rather than inventing values.
4. Capture fresh screenshots, regenerate and verify this PDF, rerun tests, and commit and push the final report and KPI SQL changes to GitHub.

Submission note: This report reflects project validation completed through 28 October 2026. The phase-2 Git tag has been pushed. The KPI target comparisons are available with documented caveats; the dashboard and KPI cards have been verified in Superset. Payroll reconciliation findings are documented separately, and unresolved differences remain disclosed for data-owner review.